

Chapter 7

Perceptual Audio Coding

The term *audio coding* usually refers to the source coding of digital audio signals, including high fidelity music, with bandwidths of up to 24 kHz. The first digital audio coding algorithms were developed in the early 1970s and used techniques such as adaptive differential pulse code modulation (ADPCM) and logarithmic quantization. These algorithms achieved typical compression ratios of 4:1 while maintaining high-quality output.

Since the early 1980s audio compression techniques have employed models of the human auditory system to achieve higher compression ratios while still maintaining good perceptual audio quality. This technique is known as *perceptual audio coding* (PAC), and a typical block diagram of an audio coder employing this technique is shown in Figure 7.1.

The *analysis* stage involves converting the time-domain signals of the sampled audio signal into frequency-domain coefficients. Because there is usually a high degree of correlation between successive audio samples, audio signals can be generally represented more efficiently in the frequency domain.

The *psychoacoustic model* stage is used to identify frequency components of the audio signal that do not contribute to the perceived quality of the reconstructed signal. These perceptually redundant components can then be encoded using fewer bits than other components that are necessary to reproduce an audio signal with a high perceptual quality.

The *dynamic quantizer* stage then uses the information produced by the psychoacoustic model to calculate the required quantizer step size for each frequency band in the audio signal. Each frequency coefficient produced by the analysis stage is then quantized using the required step size for that band.

The *bit-stream formatting* stage takes the quantized coefficients and transforms them into a binary representation. This transformation into binary data will also usually include entropy coding techniques such as Huffman coding. Any formatting overhead bits are also added at this stage such as frame headers and error correction bits.

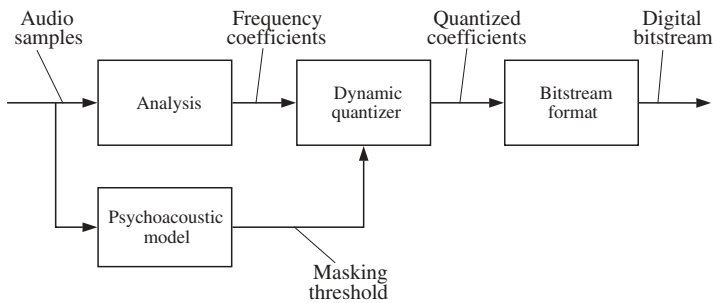


Figure 7.1 Typical block diagram of a perceptual audio coder.

The psychoacoustic model forms an integral part of any perceptual audio coder, and it is this component that is responsible for providing high-quality audio output at compression ratios of up to 12:1. In order to learn about this stage however, it is first necessary to gain a basic understanding of the human auditory system. In the remainder of this chapter, the basic anatomy and psychoacoustic properties of the human auditory system are explained.

7.1. THE HUMAN AUDITORY SYSTEM

The human auditory system can be divided into three main areas: the *outer ear*, the *middle ear*, and the *inner ear* as shown in Figure 7.2. The *pinna*, *ear canal (meatus)*, and *eardrum (tympanic membrane)* form the outer ear. Sound waves in the air travel down the ear canal and cause the eardrum to vibrate. The middle ear is an air-filled cavity that contains a group of three small bones called the *ossicles*. The ossicles

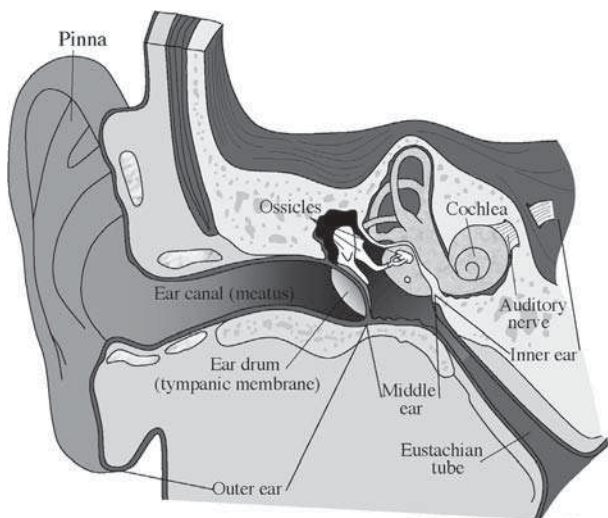


Figure 7.2 Parts of the ear.

convert the vibration of the eardrum to pressure waves in the fluid contained in the cochlea. The major components of the inner ear are the cochlea and the auditory nerve. The cochlea is responsible for transforming the mechanical vibrations of the ossicles into nerve impulses in the auditory nerve.

7.1.1. Outer Ear

The most outer part of the ear is called the pinna as shown in Figure 7.2. The shape of the pinna significantly modifies the sound waves arriving at the entrance to the ear canal. The changes in the sound waves made by the pinna depend on the direction at which they arrive. It is the shape of the pinna that allows us to determine the location of sound sources. The ear canal is a tube about 2.7 cm in length and 0.7 cm in diameter that is terminated at one end by the eardrum or tympanic membrane. The eardrum is a thin membrane containing a layer of radial fibers that form a stiff cone with an angle at its apex of about 135° .

7.1.2. Middle Ear

The individual names for the three bones that form the ossicles are the *malleus*, *incus*, and *stapes* as shown in Figure 7.3. They are more commonly called the *hammer*, *anvil*, and *stirrup* and are famous for being the smallest bones in the body. The hammer is connected directly to the eardrum at one end and makes contact with the anvil at the other. When the eardrum vibrates, this mechanical motion is passed on to the stirrup via the hammer and anvil. The head of the stirrup is connected to the anvil in a ball and socket joint and the *footplate* of the stirrup is attached to a membrane covered opening in the wall of the cochlea called the *oval window*. As

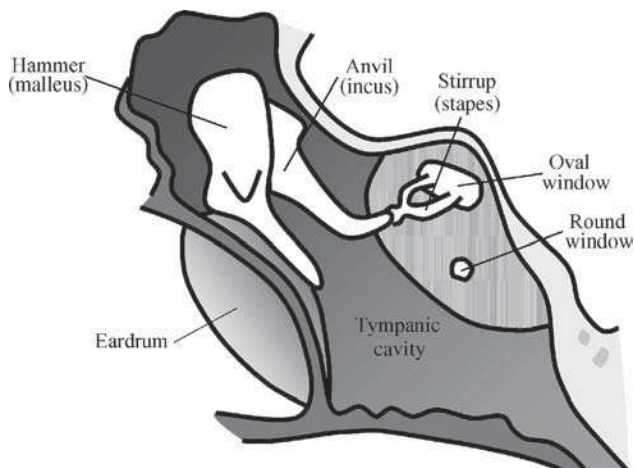


Figure 7.3 Parts of the middle ear.

the stirrup vibrates, the oval window is pushed inward and pulled outward by the footplate causing a pressure wave to be induced in the fluid of the cochlea.

The mechanical impedance of this fluid is much greater than for the eardrum, and if sound waves were to operate directly on the inner ear, they would have little effect. Because the effective area of the oval window is much smaller than that of the eardrum, the middle ear acts as a mechanical impedance matching device and allows vibrations of the eardrum to be transferred efficiently to the inner ear.

The ossicles of the middle ear also protect the inner ear from damage caused by loud sounds. The bones of the ossicles are connected to the walls of the tympanic cavity by a series of ligaments that limit their range of motion in response to vibrations in the eardrum. Once the intensity of the sound reaches a certain level, these ligaments limit the movement of the anvil, and the intensity of the pressure wave in the cochlea is also limited.

7.1.3. Inner Ear

The *cochlea* and *auditory nerve* perform the major functions of the inner ear. The cochlea forms a cone-shaped spiral of approximately two and a half turns, and the auditory nerve is intertwined throughout the center of this spiral. The main coiled structure of the cochlea is approximately 35 mm in length and has a cross-sectional area of approximately 4 mm² at the base and 1 mm² at the apex. This structure consists of two fluid-filled chambers, the *scala vestibuli* and the *scala tympani*, that are separated by the *cochlea partition*. Figure 7.4 shows a simplified illustration of the cochlea as it would look if the chambers were uncoiled. The two chambers are filled with a colorless liquid called *perilymph*. Note that the cochlea partition does not completely separate the two halves of the cochlea. There is a small opening at the apex of the cochlea partition called the *helicotrema* that allows the perilymph to flow between the two chambers.

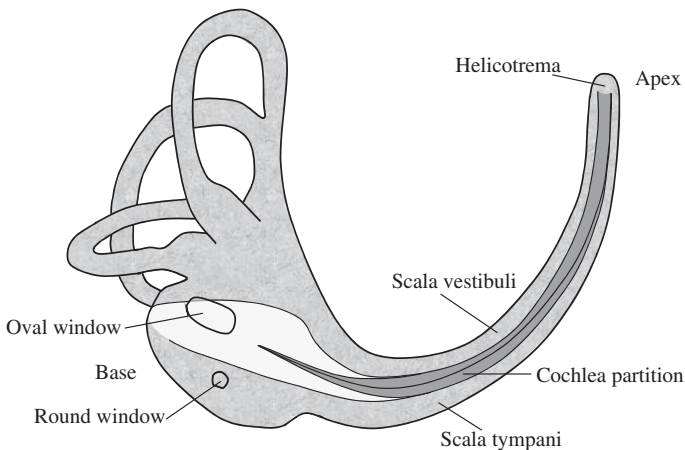


Figure 7.4 Simplified illustration of an uncoiled cochlea.

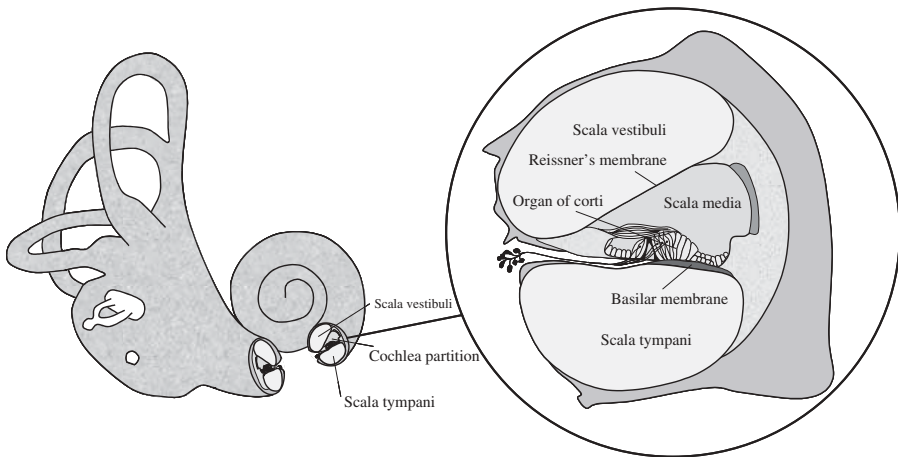


Figure 7.5 Cross section of the cochlea.

Figure 7.5 shows an expanded view of the cross section of the cochlea. The cochlea partition mainly consists of a self-contained chamber called the *scala media*. The scala media is filled with a fluid called *endolymph* and is separated from the scala vestibuli by *Reissner's membrane* and from the scala tympani by the *basilar membrane*.

The basilar membrane is attached to a complex system of hair cells and supporting structures called the *organ of Corti*. The organ of Corti contains approximately 30,000 sensory cells and is responsible for translating the mechanical movements of the basilar membrane into neural impulses. Figure 7.6 shows a more detailed view of the organ of Corti.

When the stirrup in the middle ear vibrates in response to the eardrum, the footplate induces a pressure wave in the fluid of the scala vestibuli. As this pressure wave travels down the length of the chamber, a fixed point on the cochlea partition experiences alternating levels of high and low pressure in the fluid at that location. An increase in pressure in the fluid of the scala vestibuli causes the cochlea partition to bend toward the scala tympani and hence transfer the pressure increase to the fluid in the scala tympani. This pressure increase is absorbed by the expansion of the membrane covering the *round window*. Conversely, a decrease in pressure causes the cochlea partition to bend away from the scala tympani.

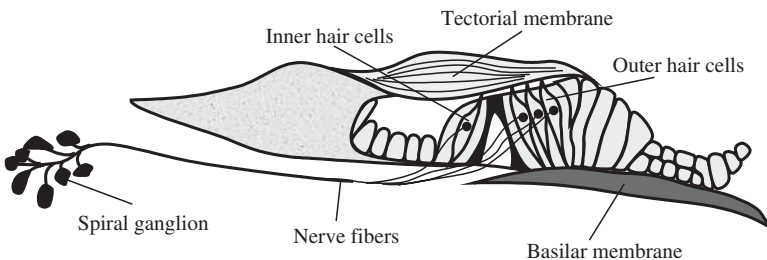


Figure 7.6 Parts of the organ of Corti.

As the cochlea partition bends in response to increasing and decreasing pressure in the scala vestibuli, the basilar membrane is displaced from its original stationary position. The displacement of the basilar membrane causes the tips of the hair cells in the organ of Corti to brush against the underside of the *tectorial membrane*. This brushing action bends the hair cells and causes nerve impulses to be sent along the auditory nerve. In this way, the inner ear converts the mechanical vibrations of the stirrup into nerve impulses that the brain interprets as sounds.

The basilar membrane is approximately 32 mm long and varies in width from approximately 0.5 mm at the apex of the cochlea to 0.1 mm at the base. As its width narrows, the basilar membrane also becomes much thicker and stiffer so that it is approximately 100 times stiffer at the base than at the apex. This variation in width and stiffness means that the basilar membrane has varying resonant properties along its length. Consequently, each point on the basilar membrane has a particular *resonant frequency*. This resonant frequency behavior can be visualized by imagining the basilar membrane as a series of very closely spaced “guitar strings” stretched across the width of the cochlear and joined together by an elastic membrane. The guitar strings at the base of the cochlear are quite short and stiff, and if one of these strings were “plucked,” it would vibrate at a high frequency. On the contrary, the guitar strings at the apex are longer and more elastic and would vibrate at a much lower resonant frequency when plucked.

For a sinusoidal pressure wave, all parts of the basilar membrane oscillate at the same frequency as the input sound. However, due to the varying resonant properties of the basilar membrane, the amplitude of the oscillations at each point along the membrane depends on the frequency. There is a point on the basilar membrane where the frequency of the input sound matches the resonant frequency of the membrane and the oscillations have maximum amplitude. The location of this point of maximum displacement is very close to the stirrup for higher frequencies and gradually moves toward the apex as the frequency decreases. Figure 7.7 shows the location of the maximum displacement for various frequencies ranging from 20 kHz to 200 Hz.

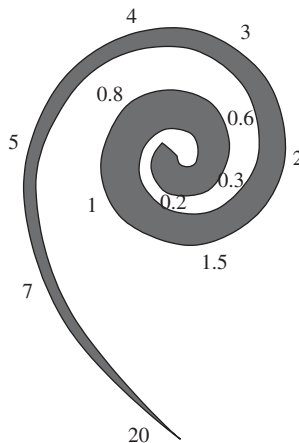


Figure 7.7 Location of the maximum displacement of the basilar membrane for various frequencies (kHz).

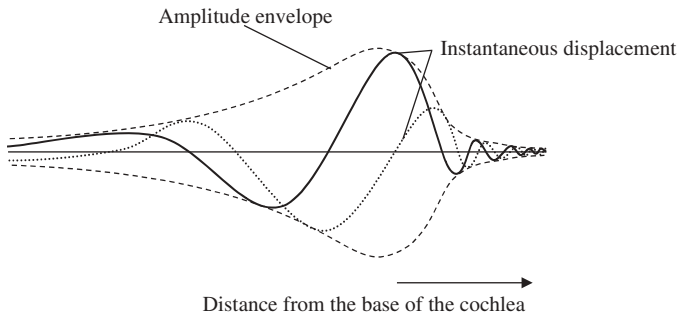


Figure 7.8 General shape of the displacement of the basilar membrane for a sinusoidal input.

As the pressure wave travels along the basilar membrane, the amplitude of the oscillations gradually increases until a maximum is reached and then rapidly decreases at points further along the cochlea. This point of maximum displacement corresponds to the point on the basilar membrane where its resonant frequency matches the frequency of the pressure wave. At this point most of the energy of the wave is absorbed by the basilar membrane, and consequently there is a rapid decrease in the amplitude of the oscillations as the wave travels further on toward the apex of the cochlea. Figure 7.8 shows the general shape of the instantaneous displacement of the basilar membrane for two successive moments in time as well as the general shape of the *amplitude envelope*, which is the line joining the amplitude peaks.

Figure 7.9 shows the instantaneous displacement of the basilar membrane for two specific cases. The double-headed arrow in each case indicates the location of maximum displacement. Figure 7.9(a) shows the displacement of the basilar membrane at one instant in time for a 5-kHz sinusoidal input, and Figure 7.9(b) shows the displacement of the basilar membrane at one instant in time for a 1-kHz sinusoidal input. Note that the point of maximum displacement is much closer to the base of the basilar membrane for the 5-kHz input than for the 1-kHz input.

The variable width and stiffness of the basilar membrane allows the cochlea to discriminate between different frequency sound waves. Sound waves of different frequencies cause different parts of the basilar membrane to oscillate with maximum amplitude, and hence the auditory nerve receives nerve impulses from different parts of the cochlea. As the frequency of an input sound varies, the brain receives nerve impulses from different parts of the auditory nerve and perceives these changes in location as variations in the *pitch* of the sound.



Figure 7.9 Instantaneous displacement of the basilar membrane for (a) a 5-kHz sinusoidal input and (b) a 1-kHz sinusoidal input.

7.2. PSYCHOACOUSTICS

The field of psychoacoustics deals with how we perceive the physical stimuli that affect the human auditory system. The air pressure waves that reach our ears are the physical stimulus, and the sound that we hear is the perceived response. The relationship between the physical properties of the stimulus and the perceptual qualities of the sound that we hear is the subject of the following sections.

7.2.1. Sound Pressure Level

A vibrating object produces a sound wave by causing air molecules to be compressed and then rarefied. As this sound wave travels past a point in space, the air pressure at this point increases and decreases at a frequency equal to the original vibrations of the sound source. Recall that pressure is a measure of force per unit area and is measured in N/m^2 . For a sinusoidal sound wave, the air pressure oscillates between a peak value above and a peak value below the ambient atmospheric pressure. The difference between these two peak values is the *peak-to-peak* amplitude of the sound wave. However, the *sound pressure* of a wave is defined as the *root-mean-square* (rms) amplitude of the sound wave, which is equal to $1/2\sqrt{2}$ times the peak-to-peak amplitude.

The human auditory system detects these pressure variations in the air. As a sound wave reaches the eardrum, the variations in air pressure cause the eardrum to vibrate, and the rest of the auditory system converts this vibration into nerve impulses that our brain interprets as sound. The human auditory system can detect a huge range of sound pressure values. The smallest sound pressure that results in an audible sound is usually given as $2 \times 10^{-5} \text{ N/m}^2$, and the loudest sound that can be tolerated has a sound pressure of 200 N/m^2 .

Because of the large range of audible sound pressure values, it is more convenient to express sound pressure values using a logarithmic scale. For this reason, sound pressure is usually converted to *sound pressure level* (SPL) and expressed in decibels using the formula in Equation (7.1).

$$\text{SPL} = 20 \log_{10} \left(\frac{P}{P_0} \right) \text{dB} \quad (7.1)$$

In Equation (7.1), P is the sound pressure and P_0 is the reference pressure value of $2 \times 10^{-5} \text{ N/m}^2$. Some examples of typical sounds and their corresponding sound pressure levels are shown in Table 7.1.

7.2.2. Auditory Thresholds

The threshold for the quietest sound that can be heard when no other sound is present is known as the *threshold in quiet* or the *absolute threshold*. Figure 7.10 shows a plot of the threshold in quiet for a typical young adult at frequencies ranging from 100 Hz to 20 kHz. This graph shows that the threshold in quiet varies appreciably with the frequency of the sound. Note that the human auditory system is most sensitive to

Table 7.1 Typical examples of sounds and their SPL values.

Example sound	SPL (dB)
Gunshot at close range	140
Jackhammer	120
Shouting at close range	100
Traffic noise	80
Normal conversation	70
Quiet conversation	50
Soft whisper	30
Watch ticking	20

sound pressure at frequencies between about 1 and 5 kHz. This sensitive range is not surprising because these are the frequencies that occur most often in human speech.

The threshold for the loudest sound that can be tolerated is known as the *terminal threshold*. This threshold does not vary significantly with frequency. A sound pressure level of 120dB causes slight discomfort, and a sound pressure level of 140dB causes a tickling sensation and pain at any frequency.

As the frequency of an input sound varies, the brain receives nerve impulses from different parts of the auditory nerve and perceives these changes in location as variations in the *pitch* of the sound. The band of audible frequencies extends from approximately 20Hz to 20kHz. Below 20Hz the variations in air pressure are too slow for the human auditory system to detect a tonal quality of the sound. Sounds above 20kHz can be heard if the sound pressure is increased to uncomfortably high levels. However, at about 23 kHz, the threshold in quiet coincides with the pain threshold, so this is effectively the upper frequency limit for hearing.

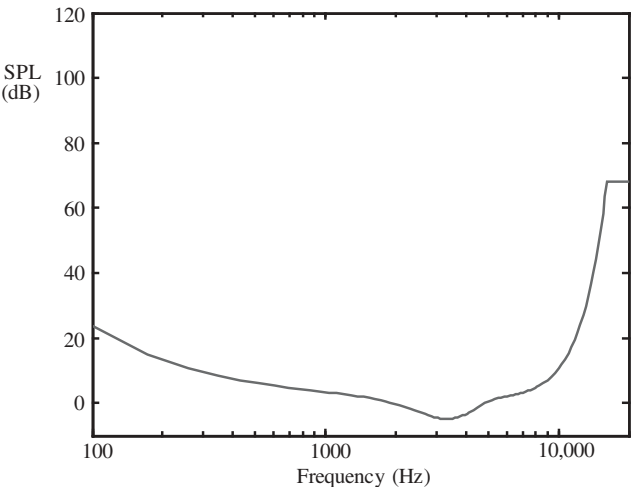


Figure 7.10 Threshold in quiet for a typical young adult.

7.2.3. The Critical Bandwidth and Auditory Filters

Early researchers observed that there was a *critical bandwidth* associated with the human auditory system at which our perception of sounds changed abruptly. Harvey Fletcher first reported this critical bandwidth in 1940 [1]. Fletcher observed the critical bandwidth by listening to a tone and band-limited noise simultaneously. To find the critical bandwidth, the frequency of a tone was set to the center frequency of the noise, and the bandwidth of the noise was then gradually widened. As the bandwidth of the noise increased, the intensity of the tone was adjusted so that it was just audible above the noise. Fletcher observed that when the bandwidth of the noise was less than the critical bandwidth, the just audible level of the tone increased as the bandwidth of the noise was increased. However, as the bandwidth of the noise was increased above the critical bandwidth, the just audible level of the tone remained the same, regardless of the bandwidth of the noise.

Another example of the critical bandwidth phenomena can be observed by listening to two tones of different frequencies simultaneously. If the frequency of the two tones differs by more than the critical bandwidth, then two distinct tones are perceived. However, if the frequency differs by less than a critical bandwidth, the sound perceived has a harsh quality, and the two distinct tones can no longer be heard.

It was also observed that the critical bandwidth is not constant for all frequencies but in fact increases with increasing frequency. Figure 7.11 shows a plot of the critical bandwidth for frequencies within the audible range.

The critical bandwidth phenomena led researchers to believe that the human auditory system acts like a bank of band-pass *auditory filters*, with continuously overlapping center frequencies. It is generally thought that the basilar membrane provides the physical basis for these auditory filters because each location on the basilar membrane has a maximum response to a particular frequency sound wave. It is also easy to see that, as the frequency is shifted above and below this resonant frequency, the amplitude of the oscillations of the basilar membrane at the original

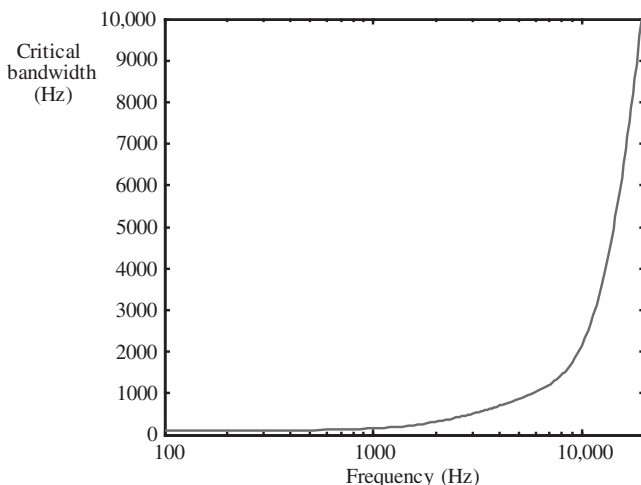


Figure 7.11 Critical bandwidth versus center frequency.

location is reduced. In this way, each point on the basilar membrane effectively acts like a band-pass filter with a different center frequency.

For all frequencies below about 16 kHz, the range of frequencies that produce significant neural activity at a fixed location on the basilar membrane corresponds to a *constant physical distance* around this fixed location. This constant physical distance determines the bandwidth of the auditory filter at each location. Remember though that, as we move along the basilar membrane from apex to base, a constant physical distance corresponds to an increasing range of frequencies. The basilar membrane therefore effectively acts like a band-pass filter bank with a continuous range of center frequencies and with the bandwidth of each filter increasing as the center frequency increases.

Empirical measurements of the critical bandwidth at various frequencies by Bertram Scharf in 1970 [2] resulted in the definition of 25 *critical bands*. The frequencies that define the center and edges of these critical bands are shown in Table 7.2. It is important to note that these critical band boundaries do not represent

Table 7.2 Center and edge frequencies for the critical bands of the human auditory system.

Band number	Lower edge (Hz)	Center (Hz)	Upper edge (Hz)
1	0	50	100
2	100	150	200
3	200	250	300
4	300	350	400
5	400	450	510
6	510	570	630
7	630	700	770
8	770	840	920
9	920	1,000	1,080
10	1,080	1,170	1,270
11	1,270	1,370	1,480
12	1,480	1,600	1,720
13	1,720	1,850	2,000
14	2,000	2,150	2,320
15	2,320	2,500	2,700
16	2,700	2,900	3,150
17	3,150	3,400	3,700
18	3,700	4,000	4,400
19	4,400	4,800	5,300
20	5,300	5,800	6,400
21	6,400	7,000	7,700
22	7,700	8,500	9,500
23	9,500	10,500	12,000
24	12,000	13,500	15,500
25	15,500	19,500	24,000

discrete nonoverlapping filters but instead represent the critical bandwidth at a set of sample frequencies.

The boundaries of the critical bands shown in Table 7.2 were then used to generate a new scale of measurement called the *critical band rate*. The unit for the critical band rate is the *Bark*,¹ and the critical band rate is sometimes called the *Bark scale*. The formula given in Equation (7.2) can be used to perform an approximate conversion from frequency to critical band rate.

$$z = \frac{28f}{f + 2200} - 0.5 \text{ Bark} \quad (7.2)$$

In Equation (7.2), z is the critical band rate and f is the frequency in Hertz. Figure 7.12 shows the relationship between the critical band rate and frequency.

The critical band rate of a sound is effectively a measure of the location on the basilar membrane that will be affected by the sound. A difference in critical band rate of less than 1 Bark indicates that two sounds are within a critical bandwidth of each other regardless of their absolute frequency. The critical band rate is also very useful in determining if one sound will affect the perception of another sound. This psychoacoustic effect is the subject of the following section.

7.2.4. Auditory Masking

Auditory masking is the general term used to describe a situation where the perception of one sound is affected by the presence of another sound. We experience forms of auditory masking regularly in everyday life. For example, when the noise from our car engine cannot be heard above the sounds from the car radio or when the

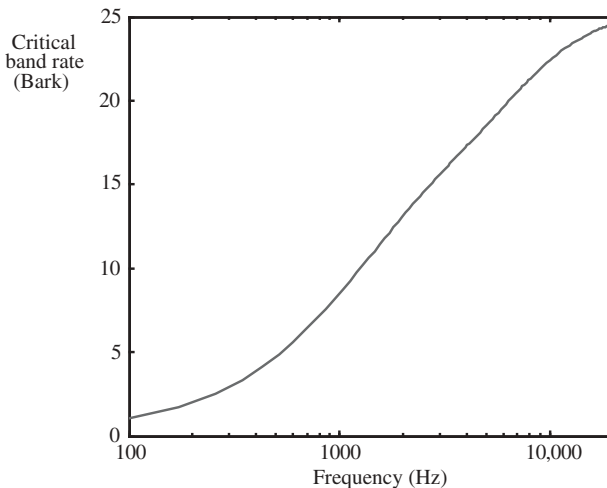


Figure 7.12 Critical band rate versus frequency.

¹The unit of Bark is named after Heinrich Georg Barkhausen, a German Physicist who introduced the phon as a unit of measurement for loudness.

music from our home stereo is inaudible because of the sound of construction work from the house next door.

In specific terms auditory masking occurs when one sound, the *masker*, raises the threshold of audibility for another sound, the *maskee*. If sound A is the masker and sound B is the maskee, then the raised threshold of sound B is called the *masked threshold* or *masking threshold*. We say that sound B has been *masked* by sound A, and the amount of *masking* is equal to the difference between the threshold in quiet for sound B alone and the masked threshold for sound B in the presence of sound A.

There are two main types of auditory masking: *frequency* masking and *temporal* masking.

7.2.4.1. Frequency Masking

Frequency masking occurs when two or more sounds are presented simultaneously, and the amount of masking depends on the difference in frequency between the masker and the maskee and the sound pressure level of the two sounds. Figure 7.13 shows a typical example of frequency masking with a 3-kHz masker and a 4-kHz maskee. The masking threshold of the 3-kHz tone indicates the sound pressure level required by a second tone before it can be heard in the presence of the masker. Because the sound pressure level of the 4-kHz tone is below the masking threshold, it is inaudible, and a human listener only hears the 3-kHz masker. For sounds that have frequencies that are quite different to the masker, the masking threshold corresponds to the threshold in quiet, and the masker has no effect on the audibility of these tones.

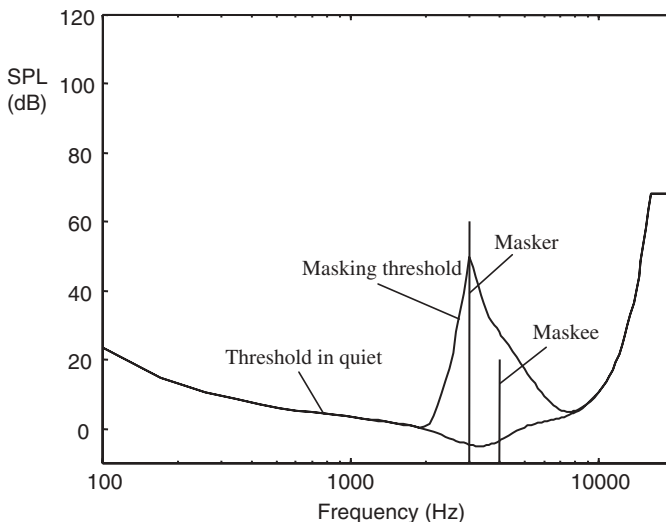


Figure 7.13 Masking threshold for a 3-kHz masker with a sound pressure level of 60 dB.

It is commonly thought that the underlying physical process involved in masking is the swamping of maskee neural activity by the masker. If the masker produces a significant amount of activity at the location on the basilar membrane that would normally be activated by the maskee, then the small amount of extra activity caused by the maskee may be undetectable.

Note that the slope of the masking threshold is less steep for frequencies above the masker frequency. This phenomenon is known as the *upward spread of masking* and is generally thought to occur because of the shape of the oscillations that are induced in the basilar membrane. Recall from Section 7.1.3 that oscillations of the basilar membrane decay away much more rapidly for parts of the basilar membrane that have resonant frequencies below the frequency of the current sound stimulus. This means that a masker tone has the best effect on parts of the basilar membrane that have a resonant frequency above the masker frequency.

In the previous section it was shown that it is the distance between the stimulated locations on the basilar membrane that will determine how one sound will affect the perception of the other. For this reason, the masking threshold for a masker is usually estimated using the critical band rate rather than absolute frequency.

7.2.4.2. Temporal Masking

Temporal masking is the term used to describe the masking effect that occurs when the masker and the maskee are not presented to the ear simultaneously. Instead, there is a small time delay between the presentation of the masker and the maskee. If the masker is presented before the maskee, this is known as *postmasking* or *forward masking*, and if the masker is presented after the maskee, it is known as *premasking* or *backward masking*. The amount of masking that occurs will depend on the time delay between the masker and the maskee, the order in which they are presented, and

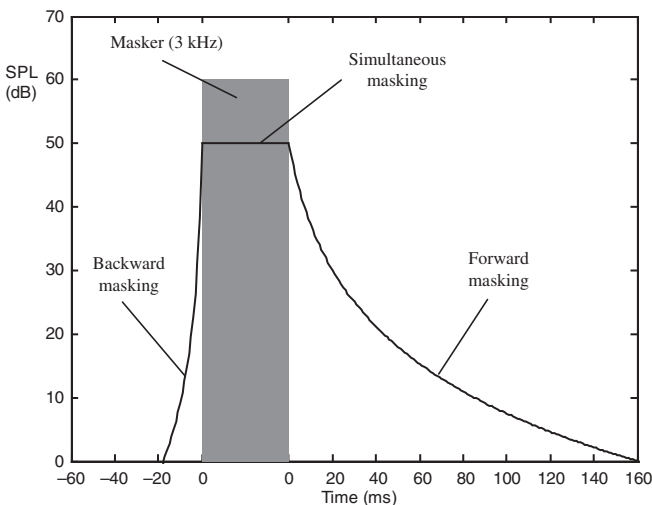


Figure 7.14 The temporal masking threshold for a 3-kHz, 60-dB masker and a 3-kHz maskee.

the difference in frequency between the two sounds. Figure 7.14 shows the temporal masking threshold for both forward and backward masking by a 60-dB, 3-kHz masker of a 3-kHz maskee.

Forward masking can occur with much longer time delays between the masker and the maskee than backward masking. Typically, backward masking cannot be observed once the maskee occurs more than 20 ms before the masker. However, forward masking can be observed with a time delay of up to 200 ms between the masker and maskee. The amount of temporal masking reduces logarithmically with the time delay between the maskee and masker. It is also interesting to observe that the maximum time delay at which temporal masking can be detected does not vary significantly with sound pressure level of the masker. As the masker sound pressure level increases, the masking threshold reduces more quickly, so the masking effect persists for the same amount of time regardless of the initial masker level. From the previous section it is also easy to see that the maximum temporal masking effect will occur when both the masker and the maskee have the same frequency.

The physical basis for temporal masking is not well understood. Two possible causes have been suggested. One is that the oscillations in the basilar membrane from the masker may not have completely decayed away before the onset of the maskee. The other suggests that recently stimulated hair cells may be reduced in sensitivity for a short time and hence emit less neural activity than would normally be the case for the maskee.

7.3. SUMMARY

In this chapter it was shown how the physical properties of the human auditory system can alter the way we perceive sounds. In particular, it was demonstrated how some sounds can mask the perception of other sounds that have a similar frequency or follow closely in time. In the following audio chapters, how these psychoacoustic limitations of the human auditory system have been exploited in the design of perceptual audio coding algorithms is explained.

PROBLEMS

1. Briefly explain the function of the following parts of the human auditory system as it converts a sound wave into neural impulses:
 - (a) Eardrum
 - (b) Ossicles
 - (c) Basilar membrane
 - (d) Hair cells
2. Explain how the variable width and stiffness of the basilar membrane allows variations in the frequency of a sound wave to be perceived as changes in the pitch of the sound.
3. If the peak-to-peak variation in air pressure for a sound wave is held constant at $4 \times 10^{-5} \text{ N/m}^2$ and the frequency is varied between 100 Hz and 20 kHz,

- (a) Determine the frequency range for which this sound will be audible to a typical young adult.
- (b) Repeat part (a) for a sound wave with a peak-to-peak variation in air pressure of $1 \times 10^{-4} \text{ N/m}^2$.

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2. B. Scharf, Critical bands, in *Foundations of Modern Auditory Theory* (J. Tobias, ed.), New York and London: Academic Press, 1970, pp. 159–202.